

Dual Space 对偶空间

History regular polyhedron

$\left\{ \begin{array}{l} \text{正四面体} \\ \text{正八面体} \end{array} \right.$ 对偶

正四面体 self-dual

$\left\{ \begin{array}{l} \text{正十二面体} \\ \text{正二十面体} \end{array} \right.$

$n=4 \leadsto 6$ 个

$n \geq 5 \leadsto 3$ 个

$$V/\mathbb{F} \quad \text{Hom}(V, \mathbb{F}) = V^* \longrightarrow \text{Hom}(V, \mathbb{F}^n) \cong (V^*)^{\times n}$$

$\dim V = n < \infty$, choose a basis $v_1, v_2, \dots, v_n$ of V

Define $v_i^* \in V^*$ by $(v_i^*)v_j = \delta_{ij} \Rightarrow v_1^*, \dots, v_n^*$ is a basis of V^*

$$\forall \sigma \in V^*, \text{ then } \sigma = \sum_{i=1}^n x_i v_i^* \quad x_i = \sigma(v_i)$$

$$\bullet \forall \sigma \in \mathcal{L}(V) \quad S^\circ = \{ \sigma \in V^* \mid \sigma(s) = 0, \forall s \in S \}$$

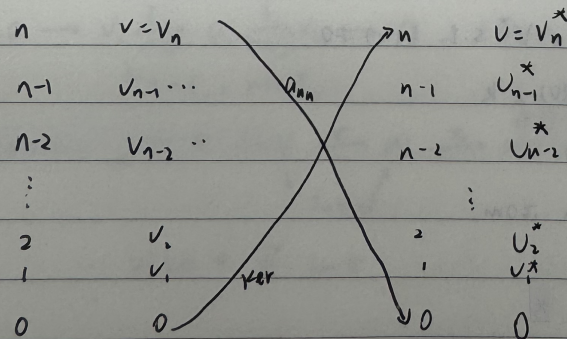
$$\bullet U \in \mathcal{L}(V^*) \quad \text{Ker } U = \bigcap_{\sigma \in U} \text{Ker } \sigma$$

Galois connection (伽罗华联络) $S_1 \subseteq S_2 \Rightarrow S_1^\circ \supseteq S_2^\circ$

$$U_1 \subseteq U_2 \Rightarrow \text{Ker } U_1 \supseteq \text{Ker } U_2$$

$\mathcal{L}(V)$

$\mathcal{L}(V^*)$



$$V \xrightarrow[\text{matrix } A]{\sigma} W$$

$$V \xleftarrow[\text{matrix } A^T]{\sigma^*} W$$

dual of σ induced by σ

$$V \xrightarrow[\text{matrix } A]{\sigma} W \quad \begin{matrix} f \circ \sigma = \sigma^*(f) \\ \downarrow \quad \quad \downarrow \\ W \xrightarrow{f} F \end{matrix}$$

Thm Let $\sigma \in \text{Hom}(V, W)$. Let A be the matrix of σ under bases β_V and β_W

Then the dual map σ^* of σ has matrix A^T under the dual bases β_W^* and β_V^*

General: $\text{Hom}(-, X) \quad V \xrightarrow{\sigma} W$

$$\text{Hom}(V, X) \xleftarrow[\text{matrix } A^T]{\sigma^* = \text{Hom}(\sigma, X)} \text{Hom}(W, X)$$

$$\sigma^*(f) = f \circ \sigma \quad \longleftarrow \quad f: W \rightarrow X$$

• $\text{Hom}(X, -) \quad V \xrightarrow{\sigma} W$

$$\text{Hom}(X, V) \xrightarrow[\text{matrix } A]{\sigma_* = \text{Hom}(X, \sigma)} \text{Hom}(X, W)$$

$$f: V \rightarrow X \quad \longmapsto \quad \sigma \circ f = \sigma_*(f): X \rightarrow W$$

Ch V. Eigen-Theory, f Linear operations

$$\text{Ex. } \mathcal{D} = \frac{d}{dx} : C^\infty \longrightarrow C^\infty$$
$$p = f(x) \longmapsto f'(x)$$

$$f(x) = p(f(x)) = \lambda f(x) \Rightarrow f(x) = ce^{\lambda x}$$

$$\text{so } \sigma(p) = \mathbb{R}$$

$$\text{c) } p = \frac{d}{dx} : \mathbb{R}[x]_n \longrightarrow \mathbb{R}[x]_n$$

$$\sigma(p) = \{0\}$$

conclusion Let $V/\mathbb{F}$ be f.d. Then $\forall f \in \text{End } V$ can be defined by $\sigma(A)$

where A is the matrix of f under any basis of V

Pf. Let $v_1, \dots, v_n$ be a basis of V

Let A be the matrix of f under this basis

$$\text{i.e. } f(v_1, v_2, \dots, v_n) = (v_1, v_2, \dots, v_n) A$$

Suppose $\lambda \in \sigma(f)$ i.e. $\exists 0 \neq v$ s.t. $\sigma(v) = \lambda v$

$$v = (v_1, \dots, v_n) x$$

$$\sigma(v) = \sigma(v_1, v_2, \dots, v_n) x = (v_1, v_2, \dots, v_n) Ax$$

$$\lambda v = (v_1, v_2, \dots, v_n) \lambda x$$

$$Ax = v x \Leftrightarrow \sigma(v) = \lambda v$$